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Abstract Book

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Preface

The **NeSS International Conference (2026)** brings together researchers, academicians, practitioners, and students from around the world to share recent advances, emerging trends, innovative methodologies, and practical applications on Statistics and Data Sciences. The conference provides a platform for exchanging ideas, promoting interdisciplinary collaboration, and fostering professional networks among participants.

This abstract book contains the accepted abstracts presented during the conference, including keynote sessions, invited talks, oral presentations, and poster presentations. The abstracts reflect the diversity of current research and demonstrate the growing impact of statistics, data science, artificial intelligence, machine learning, and quantitative methodologies across a wide range of disciplines.

The organizing committee sincerely appreciates the valuable contributions of all authors, reviewers, session chairs, keynote speakers, invited speakers, sponsors, and volunteers whose dedication and support made this conference possible. Their commitment has ensured the high academic quality of the conference programme.

We hope this abstract book will serve as a useful reference for participants during the conference and continue to be a valuable resource for future research collaborations.

Convenors

Prof. Dr. Bijay Lal Pradhan

Prof. Dr. Rabindra Kayastha

NeSS International Conference (2026)

Kathmandu, Nepal

19-20 September 2026

Message from the Conference Chair

It is my great pleasure to welcome all participants to **NeSS International Conference (2026)**, held in **Kathmandu, Nepal** from **19-20 September 2026**.

This conference provides an excellent platform for researchers, academicians, professionals, and students to present their latest research findings, exchange innovative ideas, and establish meaningful collaborations. As statistical science continues to evolve rapidly in response to emerging global challenges, meetings such as this play an important role in promoting scientific discussion and interdisciplinary research.

The scientific programme has been carefully designed to include keynote lectures, invited talks, oral presentations, and poster sessions covering a broad spectrum of topics in statistics, data science, artificial intelligence, machine learning, biostatistics, official statistics, actuarial science, and related fields.

This abstract book showcases the accepted contributions of the conference and reflects the dedication and scholarly efforts of the authors. I sincerely thank all contributors, reviewers, session chairs, members of the organizing committee, sponsors, and volunteers for their valuable support and commitment in making this conference a success.

I wish all participants a productive conference, engaging discussions, and an enjoyable stay in Kathmandu, Nepal. May this event inspire new ideas, strengthen existing collaborations, and create opportunities for future research partnerships.

Prof. Dr. Ram Prasad Khatiwada
Conference Chair
NeSS International Conference (2026)

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Contents

Preface	iii
Message from the Conference Chair	iv
Conference Committees	v
1 Keynote Speaker Abstracts	1
2 Oral Presentation Abstracts	2
3 Conference Programme	45
4 Technical Papers	47
Author Index	51

1 Keynote Speaker Abstracts

K001

Phase-wise Analysis of the Bowling Performance of Cricketers in the Nepal Premier League

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Keywords: Cricket Analytics, Data Mining in Sports, Combined Bowling Rate, Format Specific Bowling Performance Measure, Franchisee Cricket, Nepal Premier League

Cricket being a data rich sports and data management becoming simpler owing to the advances in computer technology, the sports have now become a data-driven discipline. Simultaneously, cricket has also seen mushrooming of different franchise league throughout the globe which has enhanced the need of precise player analytics. This presentation on the basis of bowling data collected from the first two seasons of the Nepal Premier League (a franchise-based Twenty20 cricket tournament established in 2024) quantifies the performance of bowlers in three distinct match phases viz. Powerplay, middle and death overs. Two different measures of bowling performance- the Combined Bowling Rate (CBR) and the Format Specific Bowling Performance Measure (FSB). The distributional pattern of the two measures were later utilized to attain practical benchmark for superlative bowlers in the different phases of a Twenty20 innings. The exercise provides the coaches with a strategic framework for the rotation of bowlers during live match and helps the franchises in making informed and cost effective bidding during player auction so that the squad has specialist bowlers for all different match situations.

2 Oral Presentation Abstracts

T001

From Anti-Spoofing to Trustworthy Audio Forensics: A Bibliometric and Science-Mapping Review of Audio Deepfake Detection

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Keywords: Anti-spoofing; Audio deepfake detection; Bibliometric analysis; Science mapping; Synthetic speech.

This study maps the intellectual structure, knowledge base, and emerging frontiers of audio-deepfake detection, addressing the absence of an integrated account of the field’s evolution, influence, collaboration, and thematic change. A PRISMA-informed bibliometric and science-mapping review analysed 945 deduplicated Scopus and Web of Science records (2020-2026; one early-online 2027 record). Performance analysis, citation and co-citation analysis, bibliographic coupling, collaboration mapping, keyword co-occurrence, thematic mapping, and trend analysis were conducted using bibliometrix/Biblioshiny. The field expanded rapidly (56.9% complete-year annual growth) but remains conference-dominated and only selectively internationalised. ASVspoof and a small set of benchmark-centred studies form the intellectual core, while recent research is shifting towards self-supervised learning, cross-dataset generalisation, real-time detection, source tracing, and explainable forensics. Persistent gaps concern multilingual and low-resource data, robustness to unseen generators and channels, privacy, fairness, and operational validation. The field is moving from benchmark optimisation towards auditable, provenance-aware, and deployment-ready detection. Progress depends on standardised cross-language evaluation, representative real-world datasets, transparent uncertainty reporting, and closer integration of technical, ethical, and governance perspectives.

Unequal Education-to-Work Pathways: Gendered Labour-Force Exclusion and SDG Interlinkages in Nepal

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Keywords: Labour-force exclusion; Education-to-work conversion; Gender inequality; Sustainable Development Goals

This study examines whether education reduces labour-force exclusion in Nepal, whether its effects differ by gender, and whether excluded adults include a discouraged but potentially employable group. Using 27,011 adults from the nationally representative Nepal Living Standards Survey 2022/23, the study applies survey-weighted descriptive analysis, logistic regression, education-by-gender interaction models, predicted probabilities, and a sub-analysis of discouraged-like exclusion. Labour-force exclusion affected 65.0% of adults. Education showed a clear threshold effect: tertiary education offered the strongest protection, while lower and intermediate levels produced limited gains. Women remained more likely than men to be excluded at every education level; even tertiary-educated women had higher predicted exclusion than men with no schooling. Among excluded adults, higher education was linked to greater willingness to work if suitable jobs were available, indicating hidden labour potential. Education alone does not guarantee access to the labor market. Nepal needs to have demand-driven vocational education system, employment opportunities, childcare provision, safe means of transport, placement service provision, and gender-aware labor market policies to conform to SDG goals 4, 5, 8, and 10. The present study merges the idea of gendered education, work transition, labor force exclusion, and discouraged labor force participation into one nationally representative approach.

An Enhanced Air Quality Index for Multi-Pollutant Air Quality Assessment

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Keywords: Air Pollution, Composite Index, Environmental Monitoring, Classification, Sensitivity Analysis

Air Quality Index (AQI) is an essential tool for communicating ambient air quality and its associated health risks. However, conventional AQI frameworks often rely on the maximum sub-index approach, which may not adequately capture the combined influence of multiple pollutants. This study proposes an improved AQI by integrating normalized pollutant concentrations using eight weighted aggregation techniques: Weighted Arithmetic Mean, Weighted Geometric Mean, Weighted Harmonic Mean, Principal Component Analysis (PCA) Index, Entropy Index, Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), Ordered Weighted Averaging (OWA), and Fuzzy Aggregation. To evaluate the robustness of these methods, a One-at-a-Time (OAT) sensitivity analysis was conducted by perturbing individual pollutant concentrations while keeping the remaining pollutants unchanged. The results revealed that all aggregation methods exhibited relatively low sensitivity to input perturbations, indicating satisfactory stability. Among them, the Weighted Harmonic Mean demonstrated the lowest average sensitivity index (0.5040%), signifying the highest robustness and reliability, whereas TOPSIS showed the greatest sensitivity to changes in pollutant values. A novel six-category AQI classification scheme, based on the the Weighted Harmonic Mean is subsequently established while preserving the familiar structure of the existing AQI models This framework offers a reliable alternative to conventional AQI methods, least effected by the extreme observations, and can support improved environmental monitoring, public health assessment, and air quality management.

Distributional and Stochastic Properties of Concomitants of Order Statistics from the Clayton Copula-Based Bivariate Gamma Distribution

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Keywords: Moment generating function; survival function; stochastic ordering; incomplete gamma function.

This paper investigates the distributional properties of concomitants of order statistics arising from a bivariate Gamma distribution constructed via the Clayton copula, referred to as the Clayton Copula-based Bivariate Gamma (CBG) distribution. Copula-based approaches offer a flexible and powerful framework for modeling dependence structures between marginal distributions. In this study, the joint and marginal probability density functions of the bivariate model, the conditional distributions, and the distribution of the r^{th} order statistic is derived. Closed-form and integral expressions for the density of the r^{th} concomitant are established, along with a finite-sum closed-form representation when the shape parameter is an integer. The moment generating function (MGF) and the first (mean) and higher order moments of the concomitant of order statistics are derived analytically. A stochastic ordering result is established, showing that higher-order concomitants are stochastically larger. The survival function of the concomitant is also obtained and analysed under varying degree of dependence. Graphical illustrations confirm the theoretical findings and highlight the versatile tail behaviour induced by the Clayton copula dependence structure.

Consumer Shopping Culture in Marts and Departmental Stores: Evidence from Ward No. 9, Kathmandu Metropolitan City

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Keywords: Shopping Culture, Consumer Behaviour, Departmental Store, Retail Marketing, Customer Satisfaction, Kathmandu Valley

The quick expansion of supermarkets and departmental stores has ended up changing the shopping culture for city consumers in Nepal, sort of in a noticeable way. Before, most people relied on small neighbourhood grocery stores, but now those kinds of practices are getting replaced more and more by organized retail outlets. These newer places provide a bigger range of goods, competitive rates, frequent promotional schemes, and what many customers feel is a better overall shopping experience.

This study is meant to look into how customers actually shop when they visit marts and departmental stores in Ward No. 9, Kathmandu Metropolitan City. More precisely, it explores how the store environment, product quality, pricing strategies, promotional activities, customer service, convenience, and technological facilities affect consumers' shopping behaviour, and also their eventual purchase decisions. This study uses a quantitative approach based on the positivist paradigm and a descriptive plus analytical research design. The primary data will be gathered from customers via a structured questionnaire, with a five-point Likert scale that keeps things consistent. Participants will be chosen using systematic random sampling, and that should subtly reduce bias. To make sense of the responses, descriptive statistics will be applied, along with reliability analysis (Cronbach's Alpha), exploratory factor analysis, correlation testing and then multiple regression analysis will also be used.

Overall, the results are anticipated to show the main drivers that shape shopping culture inside contemporary retail outlets, and then offer workable suggestions for retailers so they can improve customer satisfaction, build loyalty, and stay competitive. In addition, the research is expected to add value to the continuing body of work on consumer behaviour and organised retailing in Nepal, especially within the local context.

Comparison of the predictive performance of machine learning models for predicting low birthweight among live births in Nepal using data from the Nepal Demographic and Health Survey 2022

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Keywords: Low birth weight, Machine learning, SMOTE, NDHS

Low birth weight (LBW) is a major public health problem in Nepal and is associated with increased neonatal mortality and long-term adverse health outcomes. Machine learning (ML) offers a promising approach for early prediction of LBW, but evidence from Nepal is limited. This study compared the predictive performance of different ML models using nationally representative data. This study used data from the 2022 Nepal Demographic and Health Survey. After applying the exclusion criteria and handling missing data, 1,390 live births were included. Twenty-one(21) predictor variables were included based on literature review. Five machine learning models: logistic regression (LR), decision tree (DT), random forest (RF), support vector machine (SVM), and extreme gradient boosting (XGBoost) were developed and compared. The dataset was randomly split into training (80%) and testing (20%) sets. To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training set. Model performance was evaluated on the testing set using accuracy, sensitivity, specificity, precision, F1-score, and the area under the receiver operating characteristic curve (AUC). Predictive performance varied across the machine learning models. There was variation in the discriminative performance between the models before and after addressing class imbalance using SMOTE with RF demonstrating highest performance and SVM the greatest sensitivity. Variable importance analysis was also performed which identified variables with birth order being the most influential predictor in this analysis alongwith other factors such as geography and socioeconomic status related variables. The predictive performance of the various models varied and the effect of using SMOTE was also variable among the models. SMOTE based machine learning may support accurate detection of pregnancies with high risk of LBW.

Classical and Objective Bayesian Approaches to Estimating the Process Capability Index Cpc for the Dhillon Process Distribution

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Keywords: Process capability index; Bootstrap confidence intervals; Product spacing function, Bayesian estimation methods; Monte Carlo simulation.

The process capability index (PCI) is a valuable tool for evaluating how effectively a product meets customer expectations and for assessing overall process performance. When the quality characteristics of a process follow a normal distribution, conventional PCIs typically provide reliable and accurate results. However, these traditional indices may lead to incorrect conclusions when applied to non-normally distributed processes, potentially resulting in flawed decisions. In light of this challenge, the present study focuses on the process capability index Cpc, which remains suitable for both normal and non-normal process distributions. Specifically, we consider a process that follows the Dhillon distribution. Further, classical estimation along with classical confidence intervals and bootstrap confidence intervals for Cpc are constructed and evaluated based on their average widths and coverage probabilities. Additionally, Bayesian estimation of Cpc is carried out using both the product spacing and likelihood functions under symmetric and asymmetric loss functions assuming objective priors for the model parameters through Markov Chain Monte Carlo algorithm. A detailed Monte Carlo simulation study is conducted to evaluate the performance of the proposed estimators. Finally, the methodology is illustrated using three real datasets, thereby confirming the practicality, flexibility, and reliability of the proposed inferential framework for process capability evaluation.

Climate Stressors and Dairy Milk Production: Eastern Nepalese Farmers' Perception

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Keywords: Farmers' perception, adaptation, climate stress, disease prevalence, Koshi Province, mixed methods

Climate change has threatened livestock farming, influencing food security, rural livelihoods, and the national economy. This study assessed farmers' perception on 16 factors associated with four each from four domains: milk production, reproduction, health management, and methane production affecting dairy milk production (average milk production in litres per cattle per year). A total of 223 dairy farmers from Jhapa, Morang, and Sunsari districts in Koshi Province were surveyed in May 2024 using a structured questionnaire with a five-point Likert scale (5=very highly effect to 1= very low effect). Sex of the household head (female as a dummy variable with 1 and 0 otherwise) was also considered as a determining factor. The ordinary least square (OLS) regression models showed that the thermal stress in reproduction had the strongest positive effect on milk production, with each unit increase associated with a 0.455 increase in milk yield. Disease prevalence had the positive effect on milk production, with each unit decrease associated with a 0.319 increase in milk production. The supply of leguminous forage and concentrate on methane reduction had the positive effect on milk production, with each unit increase associated with a 0.319 increase in milk production. Results showed significant negative effect of the prevalence of vectors borne to animal health management system on milk production, with each unit increase associated with a 0.286 decline in milk production. The study presents a framework linking farmer perceptions with measured impacts and highlights the need for targeted interventions. Strengthening adaptive capacity requires improved disease prevention through vaccination, better housing to reduce heat stress, tick and vector control programs, and the development of climate-resilient leguminous forage varieties. These findings should guide extension workers, academic institutions, program planners, and policymakers in designing climate-resilient livestock strategies.

A Machine Learning-Based Framework for Tomato Maturity and Sweetness Classification in Smart Agriculture

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Keywords: Tomato quality assessment, Maturity classification, Sweetness classification, ResNet50, Principal Component Analysis, Machine learning.

Smart agriculture has emerged as a promising approach for improving crop quality assessment through the integration of artificial intelligence, computer vision, and machine learning. Horticultural crops play a vital role in human nutrition, making accurate quality assessment essential for enhancing consumer satisfaction, market value, and post-harvest management. Among these crops, tomato is one of the most widely consumed vegetables worldwide, and its quality is primarily determined by maturity stage and sweetness. Conventional methods for evaluating these quality attributes are destructive, labor-intensive, and time-consuming, limiting their suitability for rapid and non-destructive assessment. This study proposes a machine learning-based framework for simultaneous tomato maturity and sweetness classification using feature fusion. A data-set comprising 450 tomatoes and approximately 1,400 images representing three different stages of maturity, namely Light Red, Turning Red, and Ripe Red, was acquired using a smartphone camera under controlled imaging conditions. Handcrafted colour and texture features were combined with deep features extracted from a pre-trained ResNet50 model after dimensionality reduction using Principal Component Analysis (PCA) to generate a fused feature representation. Four machine learning algorithms, namely Random Forest, XG-Boost, Support Vector Machine, and Multi-layer Perceptron (MLP), were evaluated for maturity classification, where the MLP classifier achieved the highest accuracy of 91.93%. To assess internal quality, measured Brix values were grouped into Low and High sweetness categories using the k-Means clustering algorithm. The predicted maturity stage was incorporated as an additional feature for sweetness classification, resulting in a 164-dimensional feature vector, with the Random Forest classifier achieving the best accuracy of 62.46%. Furthermore, a prototype real-time tomato grading system was developed to demonstrate the practical applicability of the proposed framework. The findings demonstrate that integrating feature fusion and machine learning provides an effective non-destructive approach for tomato quality assessment and supports the development of intelligent decision-support systems for smart agriculture and post-harvest quality management.

Truncated Shanker Distribution with Properties and Applications

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Keywords: Truncation; Lifetime Distribution; Maximum Likelihood Estimation; Statistical properties; goodness of fit

Lifetime distributions play a fundamental role in reliability engineering, survival analysis, actuarial science, and biomedical research for modelling positive continuous data. However, in many practical situations, observations are available only within a restricted range due to design constraints, detection limits, or data collection procedures, making truncated models more appropriate than their parent distributions. Motivated by this need, the present study introduces the Truncated Shanker Distribution (TShD) by truncating the Shanker distribution over a specified interval. The proposed model includes left-truncated, right-truncated, and doubly truncated forms, thereby providing greater flexibility for modelling bounded lifetime data. Several statistical properties of the proposed distribution are investigated, including the probability density function, cumulative distribution function, survival function, hazard rate function, reverse hazard rate function, mean residual life function, and the first four raw moments. The model parameters are estimated using the Maximum Likelihood Estimation (MLE) method, and the likelihood equations are derived explicitly.

Quantifying the Quality of Water of Some Naturally Occurring Perennial Lakes of South Assam, India

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Keywords: Composite Index, Water Quality Index, Seasonal Variation, Sensitivity Analysis.

Wetlands are transitional zones between terrestrial and aquatic systems that remain saturated due to high groundwater or surface water for part or all of the year. Water quality is an important environmental subject and dynamic factor in wetland ecosystem productivity, which can be determined by analysing the water physically, chemically and biologically. South Assam of India, which is a valley in the foothills of the Barail Range of hills, receives enormous rainfall both pre- and during the monsoon, leading to several naturally occurring perennial lakes in the region. These lakes are the source of water for domestic use by the residents, including drinking, washing, cleaning, bathing, etc., in addition to the use of this water for irrigation. But, proper testing of the water in terms of its physical, chemical and biological properties and their changes across the different seasons of the year is rarely attempted. In this paper, the researchers attempted to quantify the water quality and their variability across the different months of the year using several composite indices for three perennial lakes of South Assam, India. The indices are then compared using sensitivity and compatibility models.

Inference on stress-strength reliability using product spacing function for the inverted exponentiated gamma distribution with application to the electrode breakdown times data

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Keywords: Inverted exponentiated gamma distribution, Stress-Strength reliability, Maximum product spacing estimation, Bayesian inference; Bayes computation

In this paper, point and interval estimation procedures for the stress-strength reliability (SSR) parameter are developed based on the inverted exponentiated gamma distribution (IEGD). The IEGD is proposed as a new and flexible lifetime distribution that exhibits desirable characteristics for modeling reliability and survival data. Classical estimation procedures for the SSR parameter are formulated using both the maximum likelihood estimation (MLE) and maximum product spacing estimation (MPSE) methods. For interval estimation, asymptotic confidence intervals (ACIs) are derived based on MLE and MPSE, along with three bootstrap confidence intervals namely standard (s-boot), percentile (p-boot), and t-bootstrap (t-boot). Furthermore, a comprehensive Bayesian framework is developed by incorporating independent gamma flexible priors under a squared error loss function, based on both the usual likelihood function and the product spacing function. Bayesian credible intervals for the SSR parameter are constructed and compared with their classical counterparts. The comparative analysis reveals that both classical and Bayesian estimators, as well as their associated intervals, exhibit satisfactory efficiency and coverage characteristics. Finally, the practical utility of the proposed methods is illustrated through a real data application, demonstrating the suitability and effectiveness of the IEGD for reliability analysis.

Do More Words Change Minds? Tracing Sentiment Transitions in Nepali Political Tweets

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Keywords: Sentiment Analysis, Political Tweets, Low resource Language

Political tweets often express opinions about political figures, parties, and events. However, it remains unclear how people form sentiment judgments when only part of a message is initially visible. This study examines how these judgments change as more words are revealed in Nepali political tweets. We use sentiment-labeled tweets from the Sentiment of Election-Based Nepali Tweets dataset, collected around Nepal’s 2022 election. Initially, 50% of the words in each tweet are randomly masked. Participants rate the visible text on a continuous sentiment scale from 0 to 5, where 0 is the most negative and 5 is the most positive. Participants then gradually reveal the masked words and may update their rating after each reveal. We record the sequence of ratings, number of reveals, time between reveals, and basic demographic information. The final analytic dataset includes 7,028 submissions from 169 participants across 1,523 unique tweets. The results show that initial sentiment judgments strongly shape later ratings. Although the average final sentiment increased by 0.32 points, the median change was 0.00, indicating substantial persistence of first impressions. Overall, 44.3% of participant-tweet pairs became more positive, 22.7% became more negative, and 33.0% remained unchanged. Compared with the original sentiment labels, the participant’s label agreement was 41.7% for the initial masked text, increasing to 52.1% after the final word was revealed; however, this did not fully remove the ambiguity. Additionally, the sentiment state transitions also became more stable after word revelation, with the same category transition probability increasing from 64.9% before any reveal to 78.0% after at least one reveal. Word-level analysis showed that masked and revealed terms were concentrated around election-related concepts such as , , , and . These findings demonstrate that progressive word reveal is a useful method for studying how partial information, early impressions, and newly revealed linguistic cues interact in human sentiment interpretation, especially for political text in low-resource languages such as Nepali.

Knowledge, Attitude and Preventive Practices Regarding Dengue Fever Among Residents of Slum Area in Butwal Sub-Metropolitan City, Rupandehi

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Keywords: dengue fever; knowledge, attitude and practice; urban slum; Nepal; ordinal logistic regression; knowledge–practice gap

Dengue fever is one of the fastest-spreading mosquito-borne viral diseases in Nepal, with urban slum communities facing increased risk because of overcrowding, unsafe water storage, and inadequate waste management. This community-based cross-sectional study assessed dengue-related knowledge, attitudes, and practices (KAP) among 401 adult residents of a slum area in Butwal Sub-Metropolitan City, Rupandehi District, Nepal, using a structured questionnaire adapted from the WHO KAP Survey Resource Pack. Data were analyzed in R using Bloom's percentage cut-offs, chi-square and Spearman's correlation tests, and logistic regression models.

Almost all participants (99.8%) had heard of dengue; however, knowledge was predominantly moderate (47.6%), with only 32.4% demonstrating good knowledge. Preventive practice was poor among 49.1% of respondents, while only 5.7% exhibited good practice. Knowledge was negatively correlated with preventive practice (Spearman's $\rho = 0.308$, $p < 0.001$), indicating that awareness did not necessarily translate into protective behaviour. Higher educational attainment was the strongest predictor of better knowledge, whereas lower educational attainment was unexpectedly associated with better preventive practices. Age, waste-disposal practices, and dengue history were significant predictors of attitude.

The findings highlight a substantial knowledge–practice gap among slum residents, suggesting that dengue control efforts should combine targeted behaviour-change interventions with improvements in water storage, sanitation, and waste-management infrastructure to promote effective disease prevention.

Multivariate Statistical Modeling for the Analysis of NEPSE Stock Index

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Keywords: NEPSE, remittance, VECM, LSTM, macroeconomic variables, forecasting.

Nepal Stock Exchange (NEPSE) plays a pivotal role in Nepal's financial ecosystem nurturing economic growth, corporate development and investment opportunity. The study of macroeconomic variables impact on the NEPSE Index is significant as it provides keen insight for investors and policy makers to make informed decisions, enhancing market efficiency and economic stability. Accurate stock price prediction remains challenging because financial market exhibits noisy, volatile, complex multifactorial nature of local and global market. Still, building a strong model is now not so far, as driven by progress in AI, access to massive data, and increased computing capabilities. This study aims to construct a predictive model for NEPSE using several new headlines to predict where the closing price would rise or descend after each one-month moment direction. Utilizing a high-density, balanced monthly longitudinal dataset spanning a 15-year horizon (2010–2025, $T = 192$), the research architecture incorporates eight critical exploratory indicators: real Gross Domestic Product (GDP), broad money supply (M2), workers' remittances, domestic consumer price inflation, central bank discount rates, the US dollar exchange rate, international gold prices, and global crude oil shocks. The Johansen trace test confirms the existence of exactly one cointegrating (long-run equilibrium) relationship between NEPSE and the eight macroeconomic variables. The estimated long-run equilibrium relationship indicates that Broad Money Supply (BMS), Gold Price, Exchange Rate, Remittance have statistically significant long-run effects on NEPSE ($p < 0.05$), whereas the interest rate, Inflation rate, GDP, Oil price does not have a statistically significant long-run effect ($p > 0.05$). The short-run dynamics indicate that only the lagged change in remittance significantly influences the current month's change in NEPSE ($p = 0.031$); the short-run effects of the remaining variables are statistically insignificant. The VECM-selected variables achieved similar forecasting performance with a simpler and more stable model, supporting a parsimonious forecasting framework. Hyperparameter tuning improved the performance of all LSTM models, yielding comparable predictive accuracy ($R^2 \approx 0.75$ – 0.80) across different variable sets.

Proportional Distribution of Childhood Cancer Types and Subtypes in Nepal: A Multicenter Hospital-Based Epidemiological Study

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Keywords: Neoplasms; Pediatrics; Epidemiology; Neoplasms by Histologic Type; Nepal

Childhood cancer is an important public health concern, yet evidence on its epidemiological distribution in Nepal remains limited. This study aimed to determine the proportional distribution of childhood cancer types and subtypes among pediatric patients in Nepal. A multicenter, hospital-based epidemiological study was conducted among children aged 18 years diagnosed with cancer at four major hospitals in Nepal during 2015-2025. Demographic and clinical data were extracted from medical records. Ethical approval was obtained from the Nepal Health Research Council (NHRC; Ref. No. 465), with permission from all participating hospitals. Descriptive statistics were used, with categorical variables summarized as frequencies and percentages and continuous variables as means and standard deviations. Among 2,180 pediatric cancer patients, leukemia was the most common cancer type (34.5%), followed by lymphoma (18.2%) and central nervous system tumors (12.7%). Cancer cases were more frequently observed among males than females, with higher proportions among children from marginalized caste communities and those residing in the Terai region. Cancer types and subtypes varied across demographic and geographic characteristics. Conclusion: Childhood cancer in Nepal shows a heterogeneous epidemiological distribution. These findings provide baseline evidence to strengthen cancer surveillance, support early diagnosis, guide pediatric oncology service planning, and promote equitable allocation of cancer care resources.

Foreign Direct Investment and Economic Growth in Nepal: Evidence from ARDL and ARIMA Forecasting Models

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Keywords: Foreign Direct Investment; Economic Growth; ARDL; ARIMA; Forecasting, MAE, RMSE

This study investigates the relationship between foreign direct investment (FDI) and economic growth in Nepal using annual time-series data from 1960 to 2024. Employing the Autoregressive Distributed Lag (ARDL) bounds testing approach, the analysis examines both short-run dynamics and long-run equilibrium relationships between GDP growth and FDI. The empirical results reveal that FDI does not exert a statistically significant effect on economic growth in the short run. However, the bounds test confirms the existence of a stable long-run cointegrating relationship between FDI and GDP growth. The estimated error correction model indicates rapid adjustment toward long-run equilibrium, suggesting that deviations caused by short-term shocks are corrected swiftly, albeit with evidence of overshooting behavior. Structural break analysis further identifies regime changes in Nepal's growth trajectory. To assess predictive performance, the forecasting accuracy of the ARDL model is compared with that of an ARIMA framework using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). The results indicate that the ARDL model provides marginally superior forecasting accuracy relative to the ARIMA model. Overall, the findings suggest that while FDI does not stimulate immediate economic growth in Nepal, its long-run contribution depends critically on domestic absorptive capacity, institutional quality, and policy stability.

Predicting the Real GDP and Its Growth Rate at Producer's Price for Next Five Years: Macroeconomic and Financial Situation

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Keywords: Real GDP, Economic Forecasting, ARIMA Model, Time Series Analysis, Monte Carlo Simulation

This study aims to forecast Nepal's Real Gross Domestic Product (GDP) growth rate at producer's price for the next five fiscal years (2025/26 to 2029/30) using the Box-Jenkins ARIMA methodology and Monte Carlo simulation techniques. Annual Real GDP data from fiscal year 1974/75 to 2024/25 obtained from Nepal Rastra Bank were analyzed. The study also examined the relationship between nominal and real GDP growth rates. Descriptive statistics, correlation analysis, linear regression, and time series forecasting methods were employed using R software. The findings reveal a moderate positive relationship between nominal and real GDP growth rates, with a Pearson correlation coefficient of 0.54. Regression results indicate that nominal GDP growth significantly predicts real GDP growth, although it explains only about 28.4% of its variation, suggesting a substantial influence of inflation and other economic factors. For forecasting, the ARIMA(2,0,0) model was identified as the most suitable model based on diagnostic tests and information criteria. The model projects a stable real GDP growth rate of approximately 4.3% to 4.4% annually over the next five years. To quantify forecast uncertainty, a Monte Carlo simulation with 2,000 future paths was conducted. The simulation estimated a 75.2% probability that real GDP growth will exceed 6% at least once during the forecast period and a 99.3% probability that real GDP will surpass 3,000,000 million rupees. The study concludes that Nepal's economy is expected to maintain moderate and stable growth in the coming years, although considerable uncertainty remains. The results highlight the importance of effective inflation management and macroeconomic stability to support sustainable real economic growth and improve the reliability of economic forecasting.

Impact of Electronic Human Resource Management Practices on Employees' Job Satisfaction in the Banking Sector

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Keywords: Commercial banks, E-HRM practices, Employees' job satisfaction, Internet and productivity

The study aims to examine the impact of electronic human resource management (e-HRM) practices, focusing on e-recruitment, e-selection, e-compensation, e-learning and training, e-performance appraisal, and e-communication, on employees' satisfaction in commercial banks in Kathmandu. Structured questionnaires were used to collect data from 401 respondents working in commercial banks in Kathmandu District. The data were analyzed using SPSS version 25. Multiple regression analysis was conducted to predict employees' satisfaction from six independent variables: e-recruitment, e-selection, e-compensation, e-learning and training, e-performance appraisal, and e-communication. The findings indicate that employee satisfaction and e-HRM practices are positively correlated. The implementation of e-HRM systems in the banking industry can encourage employee achievement, which in turn leads to increased job satisfaction. E-HRM facilitates the achievement of strategic corporate goals by enhancing employee satisfaction. E-HRM practices also create a conducive work environment in which employees develop positive attitudes toward the organization. High job satisfaction among bank employees enhances productivity, organizational citizenship behavior, and organizational commitment, while decreasing absenteeism, employee turnover, and job-related stress. Hence, a high-performing e-HRM system may be useful for commercial banks to maintain harmony between employees' personal goals and organizational goals.

Cubic Transmuted Nadarajah-Haghighi Distribution: Properties, Estimation and Application

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Keywords: Cubic transmutation map, Cubic rank transmuted Nadarajah-Haghighi distribution, Maximum product spacing, Reliability analysis, Markov chain Monte Carlo

This article suggest a new distribution, namely the cubic transmuted Nadarajah-Haghighi (CRT-NH) distribution by using cubic rank transmutation map. Various statistical properties such as pdf, reliability function, hazard function and its graphic, moment, moment generating function and order statistics of new distribution are examined. Estimation methods include maximum likelihood estimation (MLE), maximum product spacing (MPS), least squares estimation (LSE) for estimating the unknown shape, scale and transmutation parameters of the proposed model. A simulation study based on the Metropolis- Hastings Markov chain Monte Carlo (MCMC) algorithm is conducted to assess the performance of the estimators. Finally, a real data application is presented to demonstrate the flexibility and applicability of the proposed distribution.

Adaptation Strategies as Mediators of the Effects of Economic, Socio-cultural, and Environmental Drivers on Migration Intentions in Nepal

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Keywords: Migration intention, Economic, Socio-cultural, Environmental impact, Adaptation

Migration has become one of the most pressing global issues of the twenty-first century. People's migration intentions are shaped by economic, sociocultural, demographic, and environmental factors. This study investigates how perceptions of the economic, sociocultural, and environmental impacts of migration influence individuals; intentions to migrate as an adaptive response to climate change in Gandaki Province, Nepal. A survey was conducted among 493 residents in four districts (Baglung, Myagdi, Mustang, and Lamjung) of Gandaki Province to explore residents; perceptions of the drivers of migration (economic, sociocultural, and environmental) and their adaptation practices in relation to migration intention. Perceptions were collected through a structured questionnaire and measured on a five-point Likert scale. Partial Least Squares Structural Equation Modeling (PLS-SEM) was used to assess the effects of migration drivers and climate change adaptation on migration intention. This study examines whether migration intention is significantly influenced by adaptation to climate change and by the economic, sociocultural, and environmental impacts of migration. It also examines whether adaptation to climate change is influenced by the economic, sociocultural, and environmental impacts of migration. Adaptation to climate change plays a significant mediating role between environmental factors and migration intention, but not between the other drivers (economic and sociocultural) and migration intention. Hence, policy must be revised to develop effective adaptive strategies to mitigate its role in reducing people's intention to migrate.

Neurodevelopment Outcomes of Neonates with Document Sepsis – A Prospective Cohort Study

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Keywords: Neonatal Sepsis, Neuro development, Developmental Delay, Infant Health Low and Middle Income Countries (LMICs)

Neurodevelopmental outcomes in neonates are influenced by factors such as prematurity, congenital and genetic disorders, cardiac conditions, low birth weight and neonatal sepsis. Research from developing countries and low resource settings is limited. In Nepal, no such study assessing the impacts of neonatal sepsis on neurodevelopment has been reported. Understanding these impacts are crucial for early identification of high risk infants and timely interventions. This study aimed to evaluate the effect of neonatal sepsis on neurodevelopmental outcomes, provide context specific evidence to guide clinical care, follow up strategies, rehabilitation programs, and public health policies, ultimately reducing long term morbidity. The study was conducted from February 2023 to November 2025 using a prospective dual-centric cohort design at Paropakar Maternity and Women's Hospital and Siddhi Memorial Hospital. A total of 312 neonates were enrolled through non-probability purposive sampling. Neonates with documented sepsis attending the two study centers were included. Exclusion criteria comprised severe life-threatening conditions unrelated to sepsis, hypoxic-ischemic encephalopathy grade II or III, major congenital, genetic, or metabolic disorders, and extreme prematurity (<28 weeks) or birth weight <800g. Baseline sociodemographic, maternal, and neonatal data were collected. Neurodevelopmental outcomes were assessed at six months and one year after discharge using the Developmental Assessment Scale for Indian Infants (DASII). Statistical analysis was performed using SPSS v 25, applying descriptive statistics, bivariate analysis, and multivariable logistic regression to

estimate prevalence, assess associations, and identify independent predictors of neurodevelopmental delay, with $p < 0.05$ considered statistically significant. Ethical approval was taken Nepal Health Research Council. At six months follow up, neurodevelopmental assessment among 285 infants who survived neonatal sepsis showed motor delay in 20.1%, mental delay in 17.2%, combine motor and mental delay in 11.9%, and any delay in 25.3%. By one year follow up, among 269 infants, motor delay decreased to 13% and mental delay to 5.9%. Motor delay was most common in the neck control cluster at 6 months (36.1%) and in the body control cluster at 1 year (66.9%). Mental delay was highest in the memory cluster at both 6 months (22.5%) and 1 year (68.8%). At 6 months, higher odds of developmental delay were significantly associated with intrauterine growth retardation (aOR = 2.82), gestational diabetes mellitus (aOR = 3.69), postnatal steroid use (aOR = 2.49), and neonatal seizures (aOR = 2.92). At 1 year, antenatal steroid exposure (aOR = 10.08) and neonatal seizures (aOR = 4.17) were the strongest predictors of developmental delay. In this prospective cohort study nearly one-fourth of the neonates who survived documented sepsis demonstrated neurodevelopmental delay at six months. Although the proportion of affected infants declined by one year, one sixth of them continued to have developmental difficulties. Motor delays were most evident in neck and body control, while memory was notably impaired in the mental domain. Intrauterine growth restriction, maternal gestational diabetes, antenatal steroid exposure, and neonatal seizure were significant risk factors for poor outcomes. This study emphasized the importance of early screening, structured follow-up, and timely interventions for septic neonates to improve long term neurodevelopmental outcomes in Nepal.

Study of Statistical Properties of a Non-Monotonic Hazard Model: An Application to Real-Life Data from Engineering and Medical Sciences

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Keywords: Inverted U-shaped hazard; MLE; AIC; KS-test

In this paper we have introduced a new two-parameter continuous distribution with a nonmonotone hazard. Various statistical properties of the distribution, such as MLE, entropy, stochastic ordering, order statistics and ageing intensity have been studied. A simulation study has also been carried out to evaluate the performance of the point and interval estimates derived using maximum likelihood estimation. Furthermore, datasets generating out of real-life situations have also been used to assess the applicability of the distribution. The usefulness of the distribution has been checked by -2loglikelihood, AIC, CAIC and the Kolmogorov-Smirnov test, and we have found that the proposed distribution fits adequately and can be used as an alternative to some well-known distributions.

Exploring Timing of Most Recent Conception: A Data Driven Stochastic Modeling

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Keywords: Stochastic model; Timing of conception; Most recent birth; Maximum likelihood estimation

The conception stage is a critical moment in a woman's life, where insufficient attention may result in negative outcomes such as miscarriage and enduring health consequences that are a crucial driver of a nation's overall growth, necessitating vigilant monitoring of reproductive health outcomes. A significant metric in this context is the timing of birth/conception, which signifies both reproductive patterns and maternal health. The primary aim of this study is to analyze the timing of conception and thus fecundability for the most recent birth among women in Uttar Pradesh, India, utilizing data from the National Family Health Survey (NFHS) and focusing on selected indicators through some proposed stochastic models. This study assesses the waiting time to the most recent closed birth among women utilizing stochastic models such as length-biased exponential distribution (LBED), length-biased exponential gamma distribution (LBEGD) and modified mixed exponential distribution (MMED). The estimate of the parameters has been obtained through the maximum likelihood estimation procedure. The models were utilized on data from Uttar Pradesh and the Varanasi district and thereafter expanded to examine temporal trends over three rounds of the National Family Health Survey. The investigation examined demographic and socio-economic disparities, encompassing women's age, parity, sex of the most recent child, religion, caste, and residential location. The findings demonstrate that the proposed models well fit the data, revealing that over the years, the average waiting time to conception has risen, while fecundability has decreased across the majority of categories. These findings emphasize changing reproductive dynamics and reinforce the necessity for focused policy measures to enhance maternity and child health services.

Ranking Strategic Retail Items by Transaction Quality: Matched Excess Extra-Category Attached Value

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Keywords: Market basket analysis; Transaction quality; Coarsened exact matching; Average treatment effect on the treated; Empirical Bayes shrinkage; Assortment planning; Retail analytics

Merchants protect exclusives and everyday staples on the belief that those SKUs sit in better rest-of-store trips. That belief is rarely turned into an item rank. Pairwise lift cannot do it: the statistic needs a partner SKU, so a unique with no chosen consequent has no score. Ticket averages and blended indexes leave own price in the outcome and mix unlike shopping occasions, so an expensive good or a Saturday stock-up looks like quality by construction.

We take leftover revenue after the item and its subcategory come off the ticket, and compare it with trips that never carried the item but share a store cluster, week of year, weekday, and daypart. Store identity is too fine—cells empty—and basket size is the leftover itself, so neither enters the match. We call the difference excess extra-category attached value (EAV). Uniques are not interchangeable, so noisy estimates shrink toward zero rather than a peer mean. Rank is withheld unless treated volume and match rate clear preset gates. Total excess attached value (TEAV) scales EAV by trip count and must not be summed. A margin analogue (EAM) sits beside EAV and is never averaged with it. Language stays on matched association, not availability incrementality.

A stylized annual table shows why this ordering is built to disagree with sales, basket value, and lift. Lift stays the adjacency tool. The residual rank is a dollar score of rest-of-basket association that does not need a driving item. A live comparative study is left to subsequent work.

Statistical Insight and Human-Centered Data Science: Reimagining Governance, Risk, and Development Pathways in Nepal's Federal Transition

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Keywords: Human-Centered Data Science; Statistical Modeling and Governance; Climate and Risk Analytics; Federalism and Evidence-Based Policy; Ethical and Responsible AI Systems

Nepal's federal transition has unfolded alongside an unprecedented expansion in data availability, computational capacity, and analytical innovation. As governments confront complex challenges from climate vulnerability and public health disparities to administrative fragmentation and uneven development, statistics and data science have become essential tools for shaping responsive and human-centered governance. This paper explores how emerging statistical methodologies, machine learning techniques, and GenAI-driven systems can strengthen Nepal's evidence-based policymaking, while also acknowledging the social, ethical, and institutional realities that shape their application.

Drawing on examples from climate risk modeling, public health analytics, environmental monitoring, and local governance reforms, the study highlights how statistical rigor and modern data science can illuminate patterns of vulnerability, support targeted interventions, and enhance transparency across federal, provincial, and local institutions. At the same time, the paper emphasizes the human dimension of data: the lived experiences behind indicators, the inequalities embedded in datasets, and the need for interpretability and accountability in algorithmic decision-making.

The analysis argues that Nepal's development trajectory will increasingly depend on its ability to integrate statistical literacy, interoperable data infrastructures, and ethical AI frameworks into everyday governance. Bayesian modeling, spatial analytics, and machine learning offer powerful tools for small-area estimation, climate adaptation planning, and health surveillance, but their effectiveness requires institutional capacity, cross-sector collaboration, and culturally grounded implementation.

By situating Nepal's experience within global trends in statistics and data science, the paper proposes a roadmap for strengthening analytical ecosystems that are not only technically robust but also socially responsive. Ultimately, it calls for a data future where statistical insight and human-centered design work together to support equitable development and resilient gov-

ernance.

Wealth on Paper, Poverty in Pocket: A Latent Class and Ordinal Logit Analysis of Subjective Economic Well-Being in Nepal

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Keywords: Asset-Rich, Cash-Poor; Latent Class Analysis; Subjective Economic Well-Being; Liquidity Vulnerability; Ordinal Logistic Regression

Land and housing values are considered informal proxy indicators of household's wealth in Nepal, but many households in Nepal who have substantial fixed assets also struggle in the process of providing cash for their daily needs. This study examines the complexities of this "asset rich, cash poor" phenomenon, which can be hard to capture with traditional measures of poverty. The study used primary survey data from 207 households to analyze eight indicators of household financial resiliency using a technique called Latent Class Analysis (LCA): housing-status congruence, cash-flow stability, urban cost pressures, discretionary spending capacity, emergency liquidity, mandatory-expense burden, and confidence in assets as crisis insurance. The majority group (58.5%) is the Asset-Rich, Cash-Poor class, in which liquidity is constrained despite their substantial assets, reflecting economic instability; the Deprived class (14.0%) has chronic liquidity problems; the Constrained class (27.5%) has moderately constrained economic conditions. The results of the ordinal logistic regression showed that class membership was highly correlated with the self-assessed economic status. The Deprived and Constrained classes were significantly more likely to consider themselves to be poor than the Cash-Poor class ($p < .001$). Importantly, 97.5% of Asset-Rich, Cash-Poor households self-rated as "Middle" were identified as having asset-based poverty, but lacked liquidity. These findings underscore the fact that asset-based measures can mask important financial vulnerabilities among households that might be considered middle class in Nepal.

FACTORS ASSOCIATED WITH HEALTH-SYMPTOMS AMONG AGROCHEMICAL-USING FARMERS IN BHARATPUR METROPOLITAN CITY, CHITWAN

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Keywords: agrochemical, knowledge, attitude, safety practice, health-symptoms

This study examined the use of pesticides and chemical fertilizers, knowledge, attitude, safety practices, personal protective equipment (PPE) use, and self-reported health-symptoms among agrochemical-using vegetable farmers in Wards 25, 26, 27, and 28 of Bharatpur Metropolitan City, Chitwan. A cross-sectional survey design was used, and data were collected through face-to-face interviews from 110 eligible vegetable farmers selected using purposive sampling. The data were analyzed using IBM SPSS Statistics. Frequencies and percentages were used for categorical variables, while means and standard deviations were used for composite scores. Pearson's chi-square test of independence was used to examine associations between socio-demographic characteristics, knowledge, attitude, safety practice, and health-symptom level at the 5% significance level. All respondents reported using insecticides and fungicides, while 84.5% reported using herbicides. Urea and DAP were each reported by 99.1% of respondents, and potash by 93.6%. The mean knowledge score was 9.30 ± 2.03 , with 48.2% classified as having good knowledge and 51.8% as having poor knowledge. The mean attitude score was 4.91 ± 1.05 ; 66.4% of respondents had a favourable attitude and 33.6% had a less favourable attitude. The mean safety-practice score was 6.72 ± 3.09 , with 50.9% classified as having good safety practices and 49.1% as having poor safety practices. The mean health-symptom score was 4.07 ± 3.22 , with 53.6% of respondents classified as having lower symptoms and 46.4% as having higher symptoms. All socio-demographic variables showed no association with health symptoms, as the reported p-values were greater than 0.05. Knowledge was not significantly associated with health-symptoms ($\chi^2 = 3.062$, $df=1$, $p = .080$), and attitude was also not significantly associated with health-symptoms ($\chi^2 = 1.326$, $df=1$, $p = .250$). In contrast, safety practice was significantly associated with health-symptoms ($\chi^2 = 24.859$, $df=1$, $p < .001$). The findings suggest that improving safe agrochemical-handling practices should be a priority in the study area. Because the study was cross-sectional and based partly on self-reported information, the observed associations should not be interpreted as causal relationships.

SEASONALITY TRENDS IN CHILD DEATHS IN INDIA: A DIRECTIONAL STATISTICAL APPROACH

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Keywords: Child death, Seasonal Variation, Rao's spacing test, Rayleigh test

Seasonal variation in child deaths (deaths of children under five years of age) carries important implications for child health policy, yet identifying when and why deaths cluster across the calendar year requires methods that respect the cyclical nature of monthly data. This study applied circular statistical methods to examine seasonal patterns in the distribution of child deaths in India, analysing data disaggregated by state, year, religious group, and caste category using birth recode data from the National Family Health Survey (NFHS-5), covering the period 2015 to 2020. Rao's spacing test was used as the primary test of seasonal non-uniformity, supported by the Rayleigh test as a secondary measure of unimodal directional concentration. Rao's spacing test confirmed statistically significant seasonal clustering in the distribution of under-five deaths across all nineteen states, all six study years, all three religious groups, and all four caste categories examined. The Rayleigh test reached significance in only five states and two years, a systematic divergence explained by the three to five distinct monthly peaks identified in most analytical units. This multi-peak structure causes directional vector cancellation in the Rayleigh framework, artificially suppressing the mean resultant length and inflating p-values. This divergence itself is an epidemiologically meaningful finding: it confirms that under-five deaths in India are not driven by a single seasonal hazard but by the simultaneous operation of multiple disease pathways on overlapping seasonal calendars. Waterborne and vector-borne diseases including diarrhoea, malaria, and dengue drive monsoon-period peaks (June–September); acute respiratory infections dominate the winter months (December–February); heat stress and early diarrhoeal illness characterise the pre-monsoon period (March–May); and residual vector activity extends into the post-monsoon months (October–November). After accounting for the unequal lengths of calendar seasons, the apparent monsoon dominance widely reported in prior literature was shown to be a methodological artefact rather than a genuine epidemiological signal. Among social groups, Scheduled Tribes carried the highest absolute count of deaths across all seasons, reflecting compounding vulnerabilities of geographic remoteness, nutritional risk, and limited healthcare access. The anomalous pattern observed in 2020 with elevated death counts concentrated in the post-monsoon months is consistent with documented disruptions to child healthcare access and immunisation programmes during the COVID-19 pandemic. These findings indicate that child health programmes concentrated on the monsoon season alone will

miss a substantial share of preventable deaths. Year-round service delivery, with targeted intensification during identified peak periods, is more appropriate for most Indian states than seasonally focused campaigns.

STATISTICAL MODELLING OF ANXIETY LEVEL AMONG PUBLIC AND PRIVATE STUDENTS AT HIGHER SECONDARY LEVEL

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Keywords: Anxiety, Beck Anxiety Inventory, higher secondary students, institutional type, academic stream

Anxiety is an important psychological concern among higher secondary students who may experience academic demands, examination-related concerns, career expectations, and psychosocial pressures. This study aimed to assess anxiety levels and statistically model factors associated with anxiety among public and private higher secondary students of Bharatpur Metropolitan City, Chitwan. A quantitative analytical cross-sectional study was conducted among 300 Grade 11 and Grade 12 students from Science and Management streams selected using multistage sampling. Anxiety was assessed using the Beck Anxiety Inventory (BAI). Descriptive statistics, independent-samples t-test, two-way analysis of variance, bivariate analysis, and multiple linear regression were used for statistical analysis. Separate regression models were developed for public and private colleges. The mean BAI score was 19.22 ± 11.35 , ranging from 0 to 63. About 60.6% of students experienced moderate-to-severe anxiety. Public-school students had significantly higher mean BAI scores than private-school students (21.90 ± 11.32 vs. 17.06 ± 10.94 , $p = .001$), whereas the difference by academic stream was not significant ($p = 0.647$). Two-way ANOVA showed significant effects of college type ($p = 0.001$) and its interaction with academic stream ($p = 0.028$). The regression model identified fear of result, sleep duration, and female gender as significant predictors of anxiety in public schools, while birth order, fear of result, family-related factors, sleep duration, and perceived course difficulty were associated with anxiety in private schools. The findings demonstrate that anxiety is common and is associated with multiple academic, familial, and psychosocial factors, with patterns varying by institutional type.

An Extended Technology Acceptance Model (TAM) For Behavioral Intention To Adopt Digital Payment: A Study Of University Students

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Keywords: Digital payment, TAM, UTAUT, behavioral intention, university students, PLS-SEM

The rapid expansion of digital payment systems in Nepal has increased the need to understand the factors that influence their adoption intentions, particularly among university students. This study examines the determinants of university students' behavioral intention to adopt digital payment systems using an extended Technology Acceptance Model (TAM) incorporating trust, perceived risk, facilitating conditions, and subjective norms. Data were collected through a structured questionnaire from 401 university students in Nepal and analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM). The results show that 7 out of 11 proposed hypotheses were supported. Perceived ease of use significantly influences perceived usefulness and attitude toward using digital payments, while perceived usefulness also positively influences attitude. Attitude toward using digital payments has the strongest direct effect on behavioral intention. Trust positively influences perceived usefulness, perceived risk negatively affects trust, and facilitating conditions positively influence behavioral intention. However, perceived usefulness, trust, perceived risk, and subjective norms do not have significant direct effects on behavioral intention. Overall, the findings indicate that digital payment adoption among university students is a multidimensional process, with attitude, ease of use, and facilitating conditions playing particularly important roles, while trust and perceived risk primarily operate through users' perceptions rather than directly determining intention. Thus, the extended TAM provides a meaningful and useful framework for understanding digital payment adoption among university students in Nepal.

Factors Affecting Child Nutritional Status In Nepal: A Secondary Data Analysis Using Nepal Demographic & Health Survey (NDHS) 2022

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Keywords: Child malnutrition, stunting, wasting, underweight, NDHS 2022, Nepal

Child malnutrition remains a critical public health challenge in Nepal, with persistent disparities across socioeconomic and geographic groups. This study examines factors associated with stunting, wasting, and underweight among children aged 0–59 months in Nepal using child, maternal, household, and community-level characteristics. Data were drawn from the Nepal Demographic and Health Survey (NDHS) 2022, comprising a sample of 2,202 children with complete anthropometric measurements.

Nutritional status was assessed using three indicators—stunting (height-for-age), wasting (weight-for-height), and underweight (weight-for-age)—based on WHO Child Growth Standards. Descriptive statistics and Chi-square tests were employed to examine associations between selected factors and child nutritional outcomes. The findings indicate that child age, birth order, size at birth, maternal education, maternal age, maternal employment status, household wealth, sanitation facility, place of residence, and province were significantly associated with one or more forms of malnutrition.

Stunting and underweight were highest among children aged 24–35 months, whereas wasting was most prevalent among children aged 6–11 months. Children from poorer households, those with unimproved sanitation, and those living in rural areas experienced higher levels of malnutrition. Significant provincial disparities were also observed, with Karnali reporting the highest stunting prevalence and Madhesh reporting the highest wasting and underweight prevalence. In contrast, sex of the child and source of drinking water did not show statistically significant associations with any nutritional outcome. Overall, the findings underscore the multidimensional nature of child malnutrition in Nepal and highlight the importance of maternal education, improved household socioeconomic conditions and sanitation, and geographically targeted interventions in addressing child undernutrition.

Determinants Of Graduate Students' Satisfaction With Facilities And Services At The Central Departments Of IoST, Tribhuvan University

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Keywords: Student Satisfaction, SERVPERF Model, Higher Education, Graduate Students, PLS-SEM

AbstractText This Study argues that the existing frameworks of student satisfaction in higher education, while being widely applied, have a greater emphasis on material facilities and structural resources with little consideration of relational and experiential aspects in shaping student perceptions. This study uses the adapted SERVPERF model and a census-based sample of 321 third-semester master's degree students in the 13 Central Departments of the Institute of Science and Technology (IoST). The study uses Partial Least Squares Structural Equation Modeling (PLS-SEM) to assess the impact of six dimensions of service quality: Tangible, Assurance, Reliability, Responsiveness, Empathy, and Extracurricular Activities, on overall student satisfaction.

The results show that the only statistically significant predictor of student satisfaction is Empathy, which includes such sub-elements as support of the faculty, fairness, accessibility, and individual academic care. In contrast, the conventional dimensions, such as Tangibles, Reliability, and Responsiveness, do not differ significantly, indicating that in situations where infrastructural constraints are present, students are oriented toward valuable human interaction rather than physical or administrative facilities.

The findings confirm that student satisfaction is a multi-dimensional phenomenon, yet Nepalese higher education is affected less by infrastructural adequacy and more by considerable academic support, teacher-student interaction, and relational caring. Thus, this study contributes to the broader literature by challenging facility-centered models of satisfaction and illustrating the usefulness of relational quality for explaining and improving student experience.

Analyzing Factor Associated With Most Recent Closed Birth Interval Using Classification And Regression Tree Modelling Approach

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Keywords: Most recent closed birth interval, Multinomial regression, Classification and Regression Tree (CART)

Background: The study on most recent closed birth interval is of immense importance for a country like India. Among the several birth intervals, the most recent closed birth interval is determined to be more essential than others to know the recent fertility pattern. The present study was undertaken to examine the effect of several parameters like age at marriage, age at first birth, religion, caste, parity, sex of the kid, education, residence, contraceptive usage, and wealth index on Most recent closed birth interval.

Methodology: A multinomial logistic regression and Classification and Regression Tree modelling technique (CART) survival tree method are employed to accurately examine the effect of specified factors on MRCBI.

Results: Multinomial logistic regression identified age at marriage, age at first birth, present age, and survival status of prior kid as the key factors of birth interval. Women who married and started childbearing at older ages were more likely to have short birth intervals. Older women were more likely to have long birth intervals. Survival of preceding kid lowered the chances of short birth intervals and raised the chances of lengthy birth intervals. The CART model confirmed these findings and identified maternal age, reproductive history, and child survival status as the most relevant determinants of birth interval.

Conclusion: The combination of multinomial logistic regression and CART allows for a full knowledge of birth interval dynamics and can help to identify women at risk of poor birth spacing, which can in turn enable targeted maternal and child health interventions and family planning initiatives.

Statistical Distribution Of Annual Tree Growth In The Nepal Himalaya

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Keywords: Annual ring width, Dendrochronology, Gamma distribution, Himalayan forests, Lognormal distribution, Statistical modeling, Tree growth

Annual tree-ring width is a fundamental measure of radial growth in dendrochronology, but its statistical distribution should be examined before choosing a growth or climate-response model. Forests provide various ecosystem services, including fuel wood and timber, among others. Nepalese forests are affected by climate change, but there are several gaps in knowledge about the quantitative growth of many tree species in Nepal. Dendrochronology has wider applications in multiple sectors, including quantitative time-series analysis. Therefore, this study was carried out to analyze quantitative growth and its distribution in the Nepal Himalaya. By collecting tree-ring data from 63 sites across Nepal, this study analyzed the growth of 12 tree species having different age ranges from about 50 years to 1100 years. The average age of the sampled trees was 70 years, and the average growth was 2.07 mm/year. The distributional analysis of tree growth showed that the growth was right-skewed and not normally distributed.

This study demonstrates a distributional analysis using a synthetic example representing 20 trees observed annually from 1996 to 2020 ($n = 500$). Annual ring widths were generated to resemble positive, moderately right-skewed growth measurements from Himalayan conifers. Descriptive statistics and four continuous probability distributions—Normal, Lognormal, Gamma, and Weibull—were compared using maximum likelihood, Akaike's information criterion (AIC), and the Kolmogorov–Smirnov (KS) statistic. The example dataset had a mean annual ring width of 2.07 mm, standard deviation of 0.97 mm, coefficient of variation of 0.47, and skewness of 1.00.

Among the candidate models, the Gamma distribution produced the lowest AIC (1297.4), indicating the best relative fit within this illustrative comparison. The results emphasize that annual growth is non-negative and may be asymmetric, so distributional diagnostics should precede regression, time-series, or hierarchical modeling. In actual Nepal Himalaya research, the synthetic data should be replaced by crossdated field measurements, followed by appropriate detrending/standardization and assessment of temporal dependence.

The analysis provides a reproducible statistical framework for evaluating the distribution of annual tree growth and for selecting probability models suitable for subsequent climate-growth

modeling.

An Investigation Of Reproductive Complications And Intrauterine Device (IUD) Users In Uttar Pradesh, India

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Keywords: Reproductive complications, IUD users, NFHS-5, logistic regression

Objectives: Intrauterine devices (IUDs) are an invasive method of modern family planning. It is evident that the use of IUDs leads to reproductive complications such as vaginal discharge, irregular menstruation, bad odour, etc. The occurrence of reproductive complications varies across different demographic and socioeconomic groups. The purpose of this study is to examine the sociodemographic determinants of reproductive complications and use of IUDs using NFHS-5 data.

Materials and Methods: The information was collected from women surveyed in the National Family Health Survey (NFHS-5) to evaluate reproductive complications. The study used bivariate analysis and logistic regression to examine the individual and household characteristics—including maternal age, place of residence, caste, religion, wealth index, parity, and educational attainment—associated with reproductive complications among IUD users compared to the total population.

Results: Findings reveal that reproductive complications are heavily concentrated among specific vulnerable subgroups. Younger women under the age of 25 and primiparous women using IUDs experience substantially higher rates of reproductive complications compared to older and higher-parity cohorts. Socioeconomic and geographic factors also serve as strong predictors: women residing in rural areas, those belonging to marginalised castes (SC/ST), and those from the poorest wealth quintiles face a notably higher burden of complications than their urban, general-caste, and wealthier counterparts. Higher reported complications among educated women further indicate the role of symptom awareness and health literacy in reporting adverse reproductive events.

Conclusions: Results reveal that reproductive complications among IUD users are significantly influenced by maternal age, parity, wealth index, caste, place of residence, religion, and educational level. This study specifies the need for adequate pre-insertion clinical counselling, skilled provider training in rural and resource-limited settings, and routine post-insertion follow-up to improve contraceptive safety and reproductive healthcare.

People's Knowledge Of Climate Change's Impact On Tourism In The Annapurna Conservation Area, Nepal, By Developing A Climate Change Knowledge Index

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Keywords: Climate change, knowledge index, tourism, impact, Nepal

Nepal is experiencing significant environmental impacts from rising temperatures, glacial melt, and increasingly frequent extreme weather events. These changes have adversely affected public health, livelihoods, ecological systems, and the tourism sector. This study used survey data from 490 households in the Annapurna Conservation Area to assess knowledge of climate change impacts on tourism. The findings reveal that climate change awareness varies significantly by geographic location and demographic characteristics and is closely associated with age, education level, and place of residence. Urban residents exhibit notably higher awareness than their rural counterparts, and individuals with higher education demonstrate substantially greater knowledge than those with only basic education. Interestingly, literate individuals also show significantly more awareness than those with basic education. The composite climate change knowledge index yields a mean score of 62.73, indicating a moderate-to-high level of awareness among participants. While the majority possess satisfactory knowledge, approximately 42% of the population remains at low or moderate levels. These results suggest that public understanding is largely derived from experiential observations rather than formal scientific sources.

Forecasting Nepal's Nominal Gross Domestic Product Using The Box–Jenkins ARIMA Methodology

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Keywords: ARIMA, Box–Jenkins methodology, GDP forecasting, Nepal, Time series analysis, Nominal GDP

Gross domestic product (GDP) is a key indicator of economic activity and provides an important basis for macroeconomic planning, fiscal management, investment decisions, and public resource allocation. This study models and forecasts Nepal's annual nominal GDP using the Box–Jenkins autoregressive integrated moving average (ARIMA) methodology. Annual nominal GDP data at current prices for the fiscal years 1990/91 to 2024/25, comprising 35 observations and expressed in million Nepalese rupees, were analyzed using R. The GDP series was logarithmically transformed, and stationarity was assessed using graphical analysis, autocorrelation and partial autocorrelation functions, the Augmented Dickey–Fuller (ADF) test, and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test. Six pre-specified ARIMA models were estimated using maximum likelihood and compared using AIC, AICc, BIC, log-likelihood, RMSE, MAD, and residual Ljung–Box diagnostics. The results indicated that the GDP series was non-stationary in its original and logarithmic forms, while first differencing of the log-transformed series produced mixed stationarity evidence, with the KPSS test supporting stationarity. Among the pre-specified candidate models, ARIMA(1,1,2) achieved the lowest AIC and AICc and showed no significant residual autocorrelation. However, its autoregressive coefficient was estimated at the boundary of the parameter space, indicating a potential near-unit-root estimation issue. The automated model-selection procedure identified ARIMA(0,1,0) with drift as a more parsimonious specification with lower information criteria. Forecasts from ARIMA(1,1,2) indicate that Nepal's nominal GDP is expected to increase from approximately NPR 6.353 trillion in 2025/26 to NPR 7.908 trillion in 2029/30, although prediction intervals widen substantially with the forecast horizon. The findings demonstrate the usefulness of ARIMA as a transparent benchmark for Nepal's nominal GDP forecasting while highlighting the importance of model comparison, parsimony, and explicit treatment of forecast uncertainty. Future research should incorporate out-of-sample validation and compare univariate ARIMA with multivariate, machine-learning, and hybrid forecasting approaches.

3 Conference Programme

Day 1 – 19 September 2026

Time	Activity									
08:30–09:30	Registration									
09:30–10:00	Inaugural Session at CV Raman Hall									
10:00–11:00	Keynote Lecture I Title: Phase-wise Analysis of the Bowling Performance of Cricketers in the Nepal Premier League Speaker: Dibyojyoti Bhattacharjee									
11:00–11:10	Tea Break									
11:10–11:30	Invited Lecture I Title: Title Not Found Speaker: Author Not Found									
11:30–13:00	Parallel Oral Sessions									
	<table><tr><th>Area</th><th>Room</th><th>Papers</th></tr><tr><td>Statistics - Modern Data Science & AI</td><td>CV Raman Hall</td><td>T001–T006</td></tr><tr><td>Statistical Learning to Machine Learning & Big Data</td><td>Senate Hall</td><td>T007–T012</td></tr></table>	Area	Room	Papers	Statistics - Modern Data Science & AI	CV Raman Hall	T001–T006	Statistical Learning to Machine Learning & Big Data	Senate Hall	T007–T012
Area	Room	Papers								
Statistics - Modern Data Science & AI	CV Raman Hall	T001–T006								
Statistical Learning to Machine Learning & Big Data	Senate Hall	T007–T012								
13:00–14:00	Lunch Break									
14:00–15:30	Parallel Oral Sessions									
	<table><tr><th>Area</th><th>Room</th><th>Papers</th></tr><tr><td>Environment, Climate Change & Sustainability</td><td>CV Raman Hall</td><td>T025–T030</td></tr><tr><td>Emerging Trends in Bayesian Methods</td><td>Senate Hall</td><td>T031–T036</td></tr></table>	Area	Room	Papers	Environment, Climate Change & Sustainability	CV Raman Hall	T025–T030	Emerging Trends in Bayesian Methods	Senate Hall	T031–T036
Area	Room	Papers								
Environment, Climate Change & Sustainability	CV Raman Hall	T025–T030								
Emerging Trends in Bayesian Methods	Senate Hall	T031–T036								
15:30–16:00	Tea Break									
16:00–17:30	Poster Presentation Session									

Day 2 – 20 September 2026

Time	Activity		
09:00–10:00	Keynote Lecture II at CV Raman Hall Title: Title Not Found Speaker: Author Not Found		
10:00–10:20	Tea Break		
10:20–10:40	Invited Lecture II Title: Title Not Found Speaker: Author Not Found		
10:40–12:10	Parallel Oral Sessions		
	Area	Room	Papers
	New Methodologies and Recent Progress	CV Raman Hall	T049–T054
	Statistics in Risk Management	Senate Hall	T055–T060
12:10–13:10	Lunch Break		
13:10–14:40	Parallel Oral Sessions		
	Area	Room	Papers
	Public Health and Bio Statistics	CV Raman Hall	T067–T072
	Statistics in Management Sciences	Senate Hall	T073–T078
14:40–15:00	Tea Break		
15:00–16:00	Valedictory Session and Closing Ceremony at CV Raman Hall		

4 Technical Papers

ID	Paper Title	Page
T001	From Anti-Spoofing to Trustworthy Audio Forensics: A Bibliometric and Science-Mapping Review of Audio Deepfake Detection Prawesh Bhandari, Prashansha Neupane, Arju Marasini and Bishwas Sharma	2
T002	Unequal Education-to-Work Pathways: Gendered Labour-Force Exclusion and SDG Interlinkages in Nepal Arju Marasini, Prashansha Neupane, and Christopher Mbiba	3
T003	An Enhanced Air Quality Index for Multi-Pollutant Air Quality Assessment Ronit Paul, Tanusree Deb Roy, and Dibyojyoti Bhattacharjee	4
T004	Distributional and Stochastic Properties of Concomitants of Order Statistics from the Clayton Copula-Based Bivariate Gamma Distribution Ruhiteswar Choudhury and Tanusree Deb Roy	5
T005	Consumer Shopping Culture in Marts and Departmental Stores: Evidence from Ward No. 9, Kathmandu Metropolitan City Daman Bahadur Singh	6
T006	Comparison of the predictive performance of machine learning models for predicting low birthweight among live births in Nepal using data from the Nepal Demographic and Health Survey 2022 Krishna Bahadur G.C., Amit Arjyal, Haris Khurram, and Apiradee Lim	7
T007	Classical and Objective Bayesian Approaches to Estimating the Process Capability Index Cpc for the Dhillon Process Distribution Abhimanyu Singh Yadav	8
T008	Climate Stressors and Dairy Milk Production: Eastern Nepalese Farmers' Perception Dilli Ram Sedai, Nuttaya Yuangyai, Kuaanan Techato, and Basan Shrestha	9
T009	A Machine Learning-Based Framework for Tomato Maturity and Sweetness Classification in Smart Agriculture Joy Deb, Suraj Roy, and Dibyojyoti Bhattacharjee	10

Continued on next page

ID	Paper Title	Page
T010	Truncated Shanker Distribution with Properties and Applications Y. Sanjoy Kumar Singha and Rama Shanker	11
T011	Quantifying the Quality of Water of Some Naturally Occurring Perennial Lakes of South Assam, India Subham Roy and Dibyojyoti Bhattacharjee	12
T012	Inference on stress-strength reliability using product spacing function for the inverted exponentiated gamma distribution with application to the electrode breakdown times data Rajeev Kumar	13
T013	Do More Words Change Minds? Tracing Sentiment Transitions in Nepali Political Tweets Durga Pokharel	14
T014	Knowledge, Attitude and Preventive Practices Regarding Dengue Fever Among Residents of Slum Area in Butwal Sub-Metropolitan City, Rupandehi Asha Kunwar	15
T015	Multivariate Statistical Modeling for the Analysis of NEPSE Stock Index Shanti Ram Subedi and Rabindra Kayastha	16
T016	Proportional Distribution of Childhood Cancer Types and Subtypes in Nepal: A Multicenter Hospital-Based Epidemiological Study Hari Prasad Upadhyay, Krishna Sagar Sharma, Anjali Pandit, Bijay Lal Pradhan and Ram Prasad Khatiwada	17
T017	Foreign Direct Investment and Economic Growth in Nepal: Evidence from ARDL and ARIMA Forecasting Models Dipen Satyal and Bijay Lal Pradhan	18
T018	Predicting the Real GDP and Its Growth Rate at Producer's Price for Next Five Years: Macroeconomic and Financial Situation Puja Kandel	19
T019	Impact of Electronic Human Resource Management Practices on Employees' Job Satisfaction in the Banking Sector Shankar Kumar Rawat	20

Continued on next page

ID	Paper Title	Page
T020	Cubic Transmuted Nadarajah-Haghighi Distribution: Properties, Estimation and Application Hom Nath Baral, Surinder Kumar, Siddharth	21
T021	Adaptation Strategies as Mediators of the Effects of Economic, Socio-cultural, and Environmental Drivers on Migration Intentions in Nepal Madhab Prasad Baral, Vikash Kumar K. C.	22
T022	Neurodevelopment Outcomes of Neonates with Document Sepsis – A Prospective Cohort Study Dr. Swechhya Vaidya, Dr. Manu Raj, Dr. Remya Sudevan, Dr. Shailendra Bir Karmacharya, Dr. Kalpana Subedi, Dr. Ganendra Bhakta Raya, Dr. Dhurba Shrestha	23
T023	Study of Statistical Properties of a Non-Monotonic Hazard Model: An Application to Real-Life Data from Engineering and Medical Sciences Abhiraj Bhattacharya, Brijesh P. Singh	25
T024	Exploring Timing of Most Recent Conception: A Data Driven Stochastic Modeling Brijesh P. Singh	26
T025	Ranking Strategic Retail Items by Transaction Quality: Matched Excess Extra-Category Attached Value Shusan Khatiwada, Sulav Pokharel, Nayan Pandey, Bhumika Magar	27
T026	Statistical Insight and Human-Centered Data Science: Reimagining Governance, Risk, and Development Pathways in Nepal's Federal Transition Purna Bahadur Karki Ph.D.	28
T027	Wealth on Paper, Poverty in Pocket: A Latent Class and Ordinal Logit Analysis of Subjective Economic Well-Being in Nepal Govind Jnawali, Rabindra Kayastha	30
T028	FACTORS ASSOCIATED WITH HEALTH-SYMPTOMS AMONG AGROCHEMICAL-USING FARMERS IN BHARATPUR METROPOLITAN CITY, CHITWAN Samir Chapagain	31

Continued on next page

ID	Paper Title	Page
T029	SEASONALITY TRENDS IN CHILD DEATHS IN INDIA: A DIRECTIONAL STATISTICAL APPROACH Abhishek Saraswat	32
T030	STATISTICAL MODELLING OF ANXIETY LEVEL AMONG PUBLIC AND PRIVATE STUDENTS AT HIGHER SECONDARY LEVEL Himani Neupane	34
T031	An Extended Technology Acceptance Model (TAM) For Behavioral Intention To Adopt Digital Payment: A Study Of University Students Saurav Raj Paudel	35
T032	Factors Affecting Child Nutritional Status In Nepal: A Secondary Data Analysis Using Nepal Demographic & Health Survey (NDHS) 2022 Ashok Upadhyay	36
T033	Determinants Of Graduate Students' Satisfaction With Facilities And Services At The Central Departments Of IoST, Tribhuvan University Neha Rai	37
T034	Analyzing Factor Associated With Most Recent Closed Birth Interval Using Classification And Regression Tree Modelling Approach Ayushi Chourasiya, Brijesh Pratap Singh	38
T035	Statistical Distribution Of Annual Tree Growth In The Nepal Himalaya Bidur Nepal	39
T036	An Investigation Of Reproductive Complications And Intrauterine Device (IUD) Users In Uttar Pradesh, India Anushka Singha	41
T037	People's Knowledge Of Climate Change's Impact On Tourism In The Annapurna Conservation Area, Nepal, By Developing A Climate Change Knowledge Index Rajendra Prasad Upadhyaya, Vikash Kumar K.C.	42

Continued on next page

ID	Paper Title	Page
T038	Forecasting Nepal's Nominal Gross Domestic Product Using The Box-Jenkins ARIMA Methodology Udaya Bahadur Bohora, Bijay Lal Pradhan	43

Author Index

- Arjyal, Amit, 7
- Baral Hom Nath, 21
- Baral Madhab Prasad, 22
- Bhandari, Prawesh, 2
- Bhattacharjee, Dibyojyoti, 1, 4, 10, 12
- Bhattacharya Abhiraj, 25
- Bohora Udaya Bahadur, 43
- Chapagain Samir, 31
- Choudhury, Ruhiteswar, 5
- Chourasiya Ayushi, 38
- Deb, Joy, 10
- G.C., Krishna Bahadur, 7
- Jnawali Govind, 30
- K.C. Vikash Kumar, 22, 42
- Kandel Puja, 19
- Karki Purna Bahadur, 28
- Karmacharya Shailendra Bir, 23
- Kayastha Rabindra, 16, 30
- Khatiwada Ram Prasad, 17
- Khatiwada Shusan, 27
- Khurram, Haris, 7
- Kumar Rajeev, 13
- Kumar Surinder, 21
- Kunwar Asha, 15
- Lim, Apiradee, 7
- Magar Bhumika, 27
- Marasini, Arju, 2, 3
- Mbiba, Christopher, 3
- Nepal Bidur, 39
- Neupane Himani, 34
- Neupane, Prashansha, 2, 3
- Pandey Nayan, 27
- Pandit Anjali, 17
- Paudel Saurav Raj, 35
- Paul, Ronit, 4
- Pokharel Durga, 14
- Pokharel Sulav, 27
- Pradhan Bijay Lal, 17, 18, 43
- Rai Neha, 37
- Raj Manu, 23
- Rawat Shankar Kumar, 20
- Raya Ganendra Bhakta, 23
- Roy, Subham, 12
- Roy, Suraj, 10
- Roy, Tanusree Deb, 4, 5
- Saraswat Abhishek, 32
- Satyai, Dipen, 18
- Sedai, Dilli Ram, 9
- Shanker, Rama, 11
- Sharma Krishna Sagar, 17
- Sharma, Bishwas, 2
- Shrestha Dhurba, 23
- Shrestha, Basan, 9
- Siddharth, 21
- Singh Brijesh P., 25, 26
- Singh Brijesh Pratap, 38
- Singh, Daman Bahadur, 6
- Singha Anushka, 41
- Singha, Y. Sanjoy Kumar, 11
- Subedi Kalpana, 23
- Subedi Shanti Ram, 16
- Sudevan Remya, 23
- Techato, Kuaanan, 9
- Upadhayaya Rajendra Prasad, 42
- Upadhyay Ashok, 36
- Upadhyay Hari Prasad, 17
- Vaidya Swechhya, 23
- Yadav, Abhimanyu Sing, 8
- Yuangyai, Nuttaya, 9